# Mechanistic Lens on Mode Connectivity

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## Connectivity of Minimizers from a Given Dataset

- Mode Connectivity: Minimizers of modern NNs retrieved using SGD on a dataset can be connected via simple paths of zero loss.

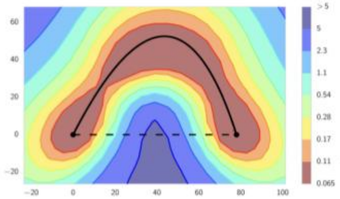
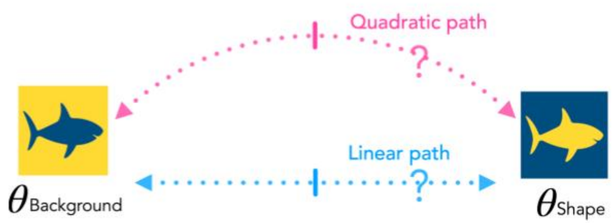


Figure: Train loss of ResNet-164 on CIFAR-100. (Garipov et al.)

Linear: $t\theta_1 + (1-t)\theta_2$;

Quadratic: $t^2\theta_1 + 2t(1-t)\theta_m + (1-t)^2\theta_2$.

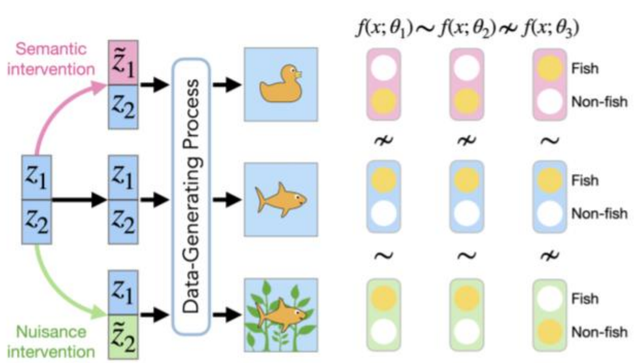
## Motivating Problem: Are Minimizers Retrieved from Different Datasets of a Task Connected?




- Q1: Are mechanistically dissimilar models, which rely on different input attributes to make predictions, connected?
- Q2: Can we exploit this connectivity?

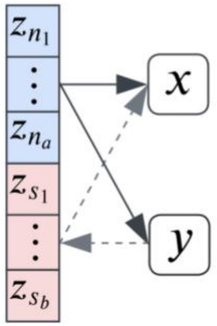
## Towards a Definition of Mechanistic Similarity

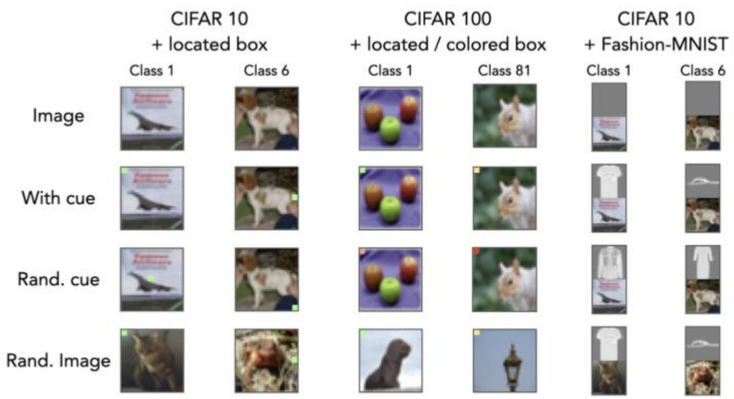
- We use shared invariances to interventions on the data-generating process to define a notion of ***Mechanistic Similarity*** between two models.




## Synthetically Inducing Mechanistically Dissimilar Models

- A data-generating process that synthetically embeds controllable, partially predictive cues in the data.
- Evaluation: Counterfactuals that break predictivity of label by altering synthetic versus natural attributes.
- Goal: Analyze connectivity of models that rely vs. ignore spurious attributes.






## Result 1: Are Mechanistically Dissimilar Minimizers Connected? Yes!

**Proposition.** ***(Mode Connectivity under Mechanistic Dissimilarity.)*** *Assume $\theta_1, \theta_2$ are minimizers of the loss on a dataset $\mathcal{D}$ and induce mechanistically dissimilar models. Given sufficient over-parameterization, there exists a continuous path along which the minimizers are mode connected.*

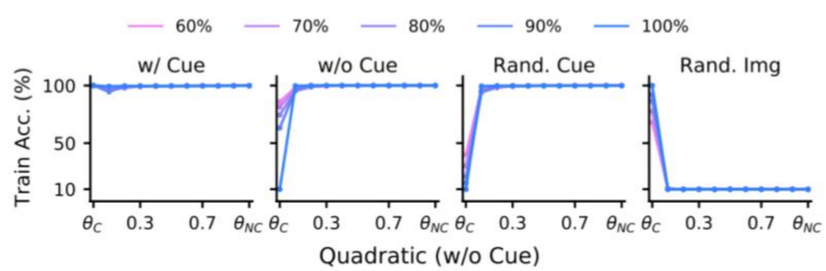



Figure: ResNet-18 models trained on located cue CIFAR-10 with different % cue data (C) show connectivity via quadratic paths to modes identified using no cue (NC) data, but are mechanistically dissimilar.

## Result 2: Is Mechanistic Similarity and Simplicity of Connectivity Paths Related? Yes!

**Conjecture.** ***(Lack of Linear Connectivity implies Mechanistic Dissimilarity.)*** *If two minimizers $\theta_1$ and $\theta_2$ of the loss $\mathcal{L}(f(\mathcal{D};\theta))$ on a dataset $\mathcal{D}$ cannot be linear mode connected (up to architectural symmetries), their induced models $f(.;\theta_1), f(.;\theta_2)$ must be mechanistically dissimilar.*

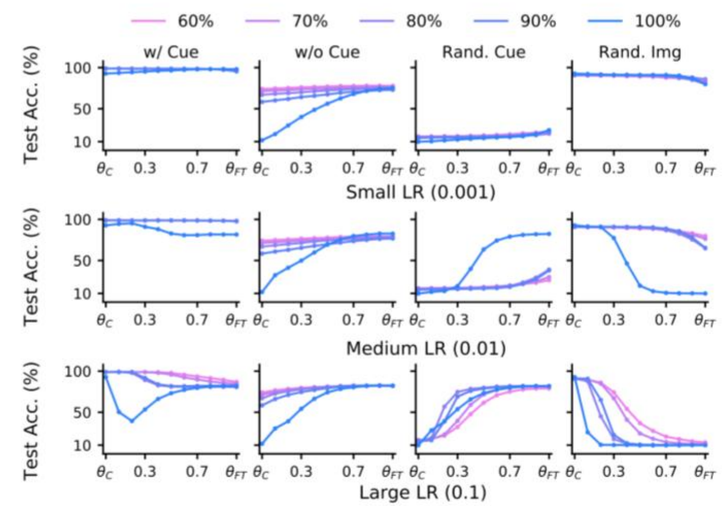
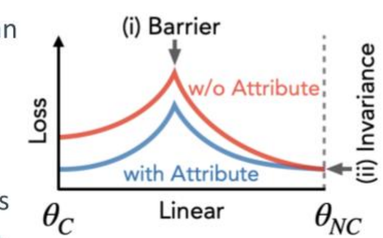



Figure: ResNet-18 models trained on located cue CIFAR-10 (C) and fine-tuned on data w/o cue (FT) show linear connectivity under small–medium learning rates, yielding mechanistically similar modes.

## Result 3: Altering Mechanisms via Lack of Linear Connectivity

- Assume a model learns to rely on an undesirable input attribute: Can we "alter" its mechanisms to use another, desirable attribute?
- Connectivity-Based Fine-Tuning (CBFT): Find a minimizer that lacks linear connectivity but is invariant to the undesirable attribute.




| | 60% Cue data | | | | 70% Cue data | | | | 80% Cue data | | | | 90% Cue data | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C-10 | NC↑ | C~ | RC~ | RI↓ | NC↑ | C~ | RC~ | RI↓ | NC↑ | C~ | RC~ | RI↓ | NC↑ | C~ | RC~ | RI↓ |
| $FT_M$ | **75.7** | 98.4 | 23.6 | 83.4 | **75.8** | 98.6 | 27.7 | 78.6 | 71.3 | 97.7 | 37.6 | 63.6 | 67.2 | 95.4 | 49.6 | 46.6 |
| $FT_S$ | 75.8 | 98.7 | 17.5 | 90.1 | 74.9 | 98.8 | 16.3 | 91.1 | 69.9 | 98.4 | 15.7 | 90.9 | 64.7 | 97.9 | 15.3 | 90.7 |
| LLR | 71.6 | 95.1 | 36.3 | 57.1 | 70.9 | 95.8 | 29.9 | 65.8 | 65.1 | 81.8 | 27.0 | 53.2 | 59.3 | 70.7 | 24.6 | 40.7 |
| LPFT | 70.6 | 88.1 | 21.0 | 70.7 | 69.6 | 87.3 | 18.7 | 72.5 | 64.4 | 63.8 | 18.8 | 48.0 | 59.7 | 56.6 | 19.8 | 37.8 |
| CBFT | 74.1 | **71.5** | **73.4** | **8.75** | 73.2 | **69.2** | **72.3** | **8.60** | 70.0 | **70.0** | **69.5** | **9.68** | **67.9** | **72.5** | **68.1** | **13.1** |